AI-Driven Multimodal Stress and Emotion Analysis with Explainable Intelligence

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Project Proposal Report

Perera K.T.K. – IT22235688

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Perera K.T.K.	IT22235688	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof. Samantha Thelijjagoda		Digitally signed by Samantha Thelijjagoda Date: 2026.03.17 11:13:55 +09'00'

ABSTRACT

Stress is a common psychological and physiological condition experienced by students and working individuals worldwide. When stress becomes prolonged, it can lead to severe health problems, including anxiety, depression, reduced productivity, and long-term physical complications. However, early stress detection is challenging because physiological responses vary between individuals, and conventional assessment methods are often based on self-reported data.

This study proposes a physiological stress detection module (C1) that utilizes wearable sensor signals to recognize stress levels with improved reliability and transparency. Signal processing is performed through filtering, sliding-window segmentation, and normalization to ensure data consistency. Relevant physiological features, including mean electrodermal activity (EDA), skin conductance peaks, heart rate, and heart rate variability (HRV), are extracted from the signals. These features are then fed into a lightweight hybrid CNN–LSTM deep learning architecture, where convolutional layers capture local temporal patterns and LSTM layers learn sequential dependencies in the physiological data.

To improve prediction reliability, probability calibration using temperature scaling is applied so that the predicted probabilities better represent actual likelihoods. In addition, the system incorporates SHAP (Shapley Additive Explanations) to enhance interpretability by identifying the physiological features that contribute most to stress predictions. It is expected that the proposed approach will produce a robust, generalizable, and interpretable stress detection model capable of generating stress probability scores along with calibrated confidence estimates and feature-level explanations. Such a system can support early stress recognition and contribute to the development of intelligent mental-health monitoring solutions.

Keywords: physiological stress detection, WESAD dataset, wearable signals, CNN–LSTM, explainable AI.

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
ECE	Expected Calibration Error
ECG	Electrocardiogram
EDA	Electrodermal Activity (skin conductance)
GSR	Galvanic Skin Response (used interchangeably with EDA)
HRV	Heart Rate Variability
LSTM	Long Short-Term Memory
ROC-AUC	Receiver Operating Characteristic-Area Under Curve
SDG	Sustainable Development Goal
SHAP	Shapley Additive Explanations

1. INTRODUCTION

1.1 Background and Context

Chronic stress is not a subjective phenomenon that is transient in nature but, in fact, it has downstream biological processes that can increase long-term health risks. Specifically, chronic stress has been linked to processes including glucocorticoid receptor resistance and inflammation, processes which have been found to be involved in more general disease risk [1]. These connections reinforce the justification of earlier diagnosis and treatment, particularly within groups under the chronic strain. At the same time, technological changes are affecting the manner in which individuals conduct their work, education and service interactions. As an example, the service domains are also becoming interested in autonomous or semi-autonomous system deployment, including frontline service robots, which can introduce new workload, interaction, and human experience patterns in service environments [2]. Practical stress monitoring in such socio-technical environments can contribute to improving human well-being in the context of changing technologies and processes.

Physiological sensing provides a more objective method for inferring stress. In the given sources, physiological signal processing is consistently placed as a technical basis that has been considered as a potential source of the determination of internal states based on measurable biosignals. Multimodal ECG and EEG signals have been suggested in stress-related work, with the application of convolutional components that are followed by sequence models to analyze patterns of temporal biosignals that are presented by deep learning architecture [6]. The proximity of physiological computing studies also shows that the EEG-based approach can be adapted to identify the state change in other contexts (e.g., fatigue when driving) which supports the overall applicability of state inference using physiological measures even when the construct of interest is not stress [7]. These two bodies of work combined encourage additional research on physiology-based stress detection, robust enough to be applied in practice.

For practical deployment, it is essential that model outputs can be regarded as decision-relevant information, and not as unqualified class labels. Within the given reference list, the concept of probability calibration has been clearly investigated within the framework of transformer-based models of detecting stress and depression in social-media text, wherein calibration has the potential to enhance the interpretation of predicted probabilities as likelihood-like quantities to make a downstream decision [9]. Although that work is not a physiology-based, it gives a useful design indication to stress-related AI systems: when downstream decision processes are likely to consume predicted values, it should be treated as an output that may require calibration. Last, physiological stress measurement is by very nature sensitive biometric data. The Body-area Networks Research: Use of schemes to key agree based on multi-biometric and physiological signals has been proposed, and it is important to note that the secure usage and reliable treatment of physiological information is indeed an active issue, when the information is moved or manipulated within a network [10]. All this puts the issue in perspective: there is a strong impulse to the development of a physiology-grounded stress-detection module that integrates the plausible biosignal modeling with practically deployable probability results and accountable biometric data processing [1], [6], [9], [10].

1.2 Literature Review

In the given references, studies relevant to AI-based stress detection include physiological modeling, multimodal learning, probability calibration, and biometric system support technologies. One such stress-oriented contribution in the reference set is the proposal of multimodal stress detection based on ECG and EEG signal analysis, with explicit use of deep learning models such as convolutional neural networks (CNN) and long short-term memory (LSTM) networks to identify patterns in biosignal time series [6]. This provides a clear direction, suggesting that multimodal biosignal fusion combined with temporal deep learning can be used for stress inference under the experimental conditions of such studies [6].

In another related domain, complementary physiological signal research demonstrates similar methodological capabilities. For example, Li proposes an EEG-based recognition model for detecting driving fatigue, showing that EEG-derived features and machine learning models can classify physiological or cognitive states in practical applications, even when the target construct is not stress [7]. From a methodological perspective, this indicates that physiological signals contain discriminative information about human states, and that pipelines involving preprocessing, feature extraction, and classification can be adapted across domains, although stress-specific validation and construct clarity are necessary [7].

Meanwhile, the reference set also includes work focusing on probability calibration in stress-related classification tasks. Ilias, Mouzakitis, and Askounis investigate transformer-based model calibration for detecting stress and depression in social media text [9]. The significance of this work lies in its emphasis on the reliability of predicted probabilities rather than only classification accuracy. This is particularly important when model outputs are interpreted as estimates of risk or confidence that may influence downstream decisions [9]. Although the modality in this case is textual rather than physiological signals, the concept of probability calibration highlights the potential gap between raw model outputs and decision-ready probabilities, which can be relevant to stress detection systems more broadly [9].

The references also include studies related to emotion recognition and multimodal affective computing. Kang et al. propose a modality-adaptive emotion recognition system, emphasizing that multimodal affect recognition may require adaptation across different modalities and contexts [3]. Similarly, Livingstone and Russo introduce the RAVDESS dataset, a multimodal database containing vocal and facial expressions in North American English that is widely used in emotion recognition research [11]. While these studies contribute to the broader field of affective computing, they also highlight a conceptual limitation: emotion recognition datasets do not necessarily provide validated stress labels, and emotional categories are not always equivalent to stress as a psychological construct [3], [11].

Finally, the reference set also includes technologies related to biometric security and broader machine learning applications. Reshan et al. present a multi-biometric and physiological signal-based key agreement protocol for body-area networks, demonstrating that secure processing and management of physiological data is an important concern in wearable systems [10]. In another area of multimodal machine learning, Wood et al. present a high-fidelity gaze-redirection method in video, illustrating the complexity that can be achieved in temporal and perceptual modeling, even though the work is not directly related to stress detection [8]. Taken together, these studies demonstrate the feasibility of multimodal modeling, the importance of probability calibration, and the need for secure handling of biometric data. However, the integration of these elements into a physiology-based stress detection module that is both reliable and responsibly deployable remains an open research challenge [6], [9], [10].

1.3 **Research Gap**

The synthesis of the referenced studies reveals a clear integration gap rather than a complete absence of relevant methods. On one hand, multimodal physiological stress detection using deep learning has already been proposed. For example, StressNet integrates ECG and EEG signals with CNN and LSTM components to learn temporal representations for stress detection [6]. On the other hand, probability calibration has been studied as an important methodological issue in stress-related classification tasks, particularly in detecting stress and depression from social media text. In that work, calibration techniques are applied to improve the reliability of predicted probabilities used in decision-making processes [9]. However, while stress-related AI models in other modalities explicitly incorporate probability calibration [9], there is limited emphasis on calibrated probability outputs as a standard component in multimodal physiological stress detection systems [6]. This highlights a research opportunity to integrate (i) multimodal physiological stress inference and (ii) explicit probability calibration as a key design objective for deployable stress detection models [6], [9].The second gap relates to construct alignment and dataset suitability in studies situated within the broader field of affective computing. The reference list includes research on emotion recognition systems and datasets, such as modality-adaptive emotion recognition models and multimodal emotion databases [3], [11]. While these studies contribute valuable insights into multimodal affect modeling, they also reveal a conceptual limitation: emotion recognition datasets do not necessarily provide validated stress labels or clearly defined stress constructs. As a result, emotion recognition models cannot always serve as reliable proxies for stress detection [3], [11]. Therefore, a stress detection system must clearly define stress as the target construct and ensure that the learning problem is designed and evaluated using datasets that explicitly represent stress-related physiological responses.

The third gap is methodological and system-level, particularly concerning model evaluation, generalization, and responsible deployment. Larsson highlights that observational research must carefully consider study design characteristics when interpreting long-term outcomes, as design decisions can significantly influence the validity of conclusions [5]. In the context of physiological stress detection, this implies the need for robust evaluation strategies that reduce overly optimistic performance estimates and improve generalizability across individuals and contexts. Furthermore, physiological stress detection relies on sensitive biometric data, making security and reliability important considerations. Research on physiological-signal-based key agreement protocols for body-area networks demonstrates that secure processing and protection of physiological data remain active technical challenges in wearable systems [10]. However, the existing literature does not clearly explain how multimodal physiological modeling, probability calibration, and secure biometric data handling can be integrated into a unified stress detection system suitable for real-world deployment [6], [9], [10].

Overall, the main research gap addressed in this study is the lack of integrated approaches for physiology-based stress detection that combine multimodal biosignal modeling [6], probability calibration techniques inspired by stress-related AI in other modalities [9], robust evaluation strategies for generalization [5], and responsible handling of sensitive biometric data in wearable environments [10].

2. OBJECTIVE

2.1 Main Objective

To create and deploy an explainable subject-independent physiological stress-detection module based on wearable sensor signals that generates calibrated stress probabilities and the capability to explain features.

2.2 Specific Objectives

(i). **Data collection and pre-processing:** Gather open wearable datasets (WESAD - Wearable Stress and Affect Detection from Kaggle) and carry out signal pre-processing (including notch and bandpass filtering) and segmentation into fixed 5-second windows. Normalize signals across subjects.

(ii). **Feature engineering:** Obtain Physiological features (from EDA (mean, phasic response amplitude, number of skin conductance peaks) and from ECG (heart-rate, heart-rate variability).

(iii). **Model development:** Train a lightweight CNN–LSTM stress classifier that combines EDA and ECG features, with fewer than 1 million model parameters at inference time, to support real-time prediction.

(iv). **Probability calibration:** Calibrate the raw model output using temperature scaling and compute the Expected Calibration Error (ECE).

3. METHODOLOGY

Research Approach

The project follows a System Development and Evaluation research methodology, which integrates design science principles with controlled experimentation. This approach is appropriate for research where a new or improved system is developed, and its performance is evaluated against baseline methods. In this study, the proposed stress detection module will be developed as a research artefact and quantitatively evaluated to demonstrate improvements in stress detection accuracy, probability calibration, and model explainability.

Data Collection and Participant Selection

Publicly available wearable datasets, primarily the WESAD (Wearable Stress and Affect Detection) dataset, will be used as the main data source. To manage computational complexity, a subset of approximately 8–10 participants will be selected for experimentation. A subject-independent data split will be applied so that no individual appears in both the training and testing sets. This approach prevents data leakage and ensures that the model evaluation reflects its ability to generalize to unseen users. Since only publicly available and de-identified datasets are used, privacy and ethical concerns are minimized.

Data Pre-processing

Physiological signals require several preprocessing steps before they can be used for machine learning.

Signal Cleaning:

Band-pass filtering will be applied to remove sensor-specific noise, while notch filtering will be used to reduce power-line interference. In addition, signal artefacts and extreme outliers will be detected and removed.

Windowing:
Continuous EDA and ECG signals will be segmented into fixed-length windows (e.g., 5-second windows) so that each segment can be treated as an independent training sample.

Normalization:
Z-score normalization will be applied at the participant level to reduce inter-subject variability and ensure consistency across signals.

Feature Engineering

Physiological features will be extracted from each segmented window.

From EDA signals, the following features will be extracted:

- Mean skin conductance
- Phasic response amplitude
- Number of skin conductance peaks

From ECG signals, the following features will be computed:

- Heart rate
- Heart rate variability (HRV)
- Statistical features of RR intervals

These extracted features will be combined to form a feature vector that will be used as input to the stress detection model.

Model Development

A lightweight CNN-LSTM hybrid architecture will be developed for stress classification. The convolutional layers will learn local temporal patterns from the physiological signals within each window, while the LSTM layers will capture longer-term sequential dependencies across time. The final dense layer will output the predicted stress probability.

To ensure computational efficiency and practical usability, the model will be designed with fewer than one million parameters, enabling near real-time inference on consumer-grade hardware.

Explainability and Probability Calibration

To improve interpretability, Shapley Additive Explanations (SHAP) will be used to explain model predictions by quantifying the contribution of each physiological feature to the predicted stress level. This allows feature-level explanations for individual predictions and supports participant-level interpretability.

In addition, temperature scaling will be applied as a probability calibration technique. This method adjusts the model's output probabilities so that they better reflect the true likelihood of stress predictions.

Evaluation Strategy

Model performance will be evaluated using a subject-independent train-test split. The proposed CNN-LSTM model will be compared with baseline classifiers such as logistic regression or support vector machines (SVM) trained on the same feature set.

The following evaluation metrics will be used:

- Classification accuracy
- ROC-AUC score
- Confusion matrix analysis

To assess probability calibration, the Expected Calibration Error (ECE) will be calculated. Experiments will be repeated across multiple random splits, and statistical significance will be tested using paired t-tests.

In addition, inference time per window will be measured to evaluate the model’s ability to operate in real-time applications.

Tools and Implementation

The experiments will be implemented in Python using scientific computing libraries including NumPy, SciPy, pandas, and PyTorch. Model training and evaluation will be performed in a controlled cloud environment such as Google Colab Pro. The final trained model may be exported using formats such as ONNX for integration into the broader multimodal stress-emotion analysis system.

Ethical Considerations

This study uses only publicly available, de-identified datasets, and therefore does not involve the collection of new personal data. All analyses will follow standard privacy and confidentiality practices. The reported results will not reference individual participants, ensuring ethical compliance in data usage and reporting.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

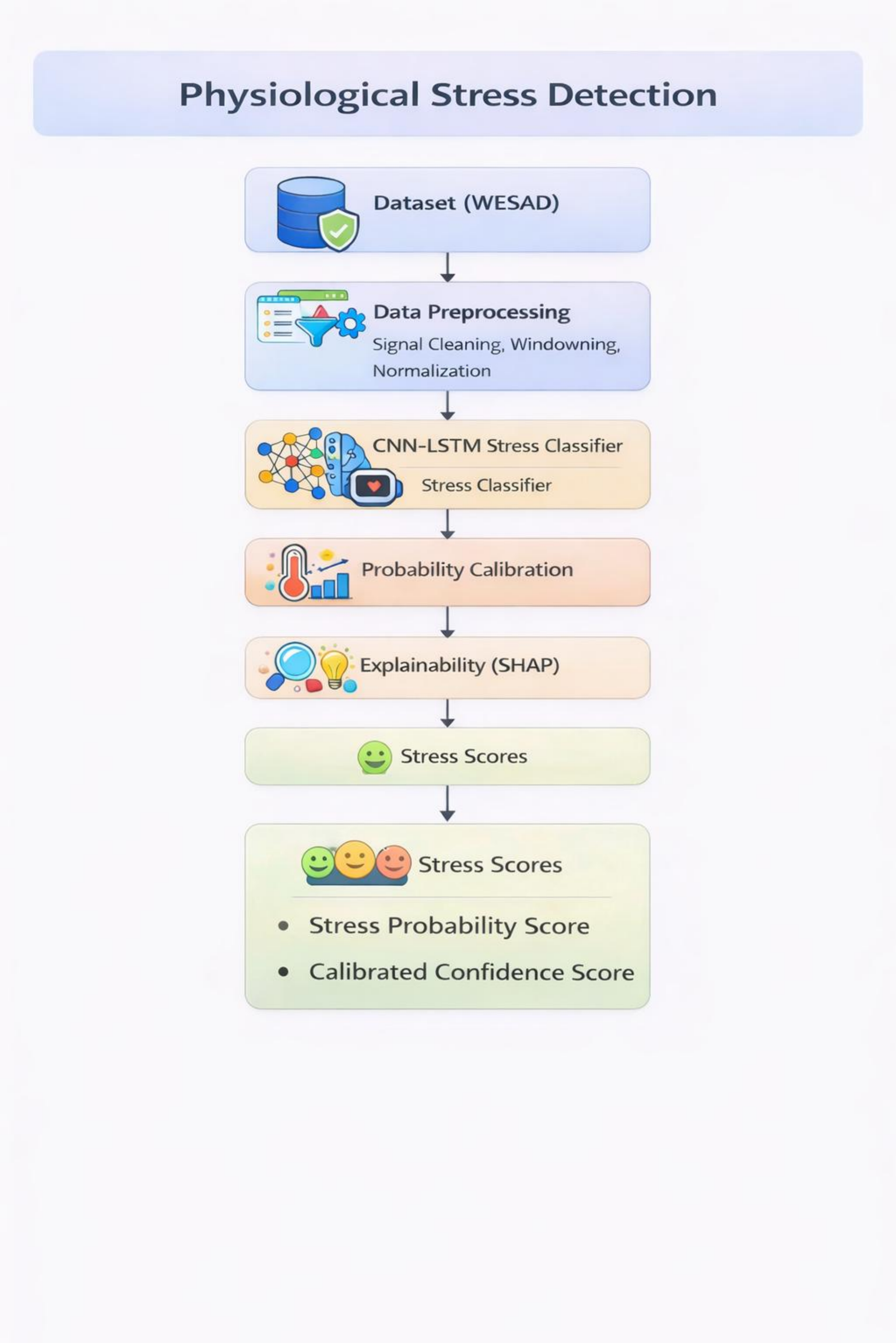

Figure 1- High-Level System Architecture

5. USER REQUIREMENTS

The main stakeholders include people who are interested in tracking stress (students, working professionals), experts in mental health applications and developers of digital wellbeing tools. Requirements are gathered from the problem description and validated with domain experts.

Functional requirements

No.	Requirement
FR1	Load and manage physiological signals (EDA and ECG) from the WESAD dataset for stress analysis.
FR2	Preprocess physiological signals using cleaning, band-pass filtering, artifact removal, window segmentation (5-second sliding windows), and z-score normalization.
FR3	Extract physiological features from EDA and ECG signals including mean EDA, phasic response amplitude, number of skin conductance peaks, heart rate, HRV, and RR-interval statistics.
FR4	Classify each signal window using a CNN–LSTM model to generate a stress probability score.
FR5	Apply probability calibration (Temperature Scaling) to ensure reliable and well-calibrated stress probability predictions.
FR6	Generate explainable AI outputs using SHAP to identify dominant physiological features contributing to stress predictions.
FR7	Provide structured outputs including stress probability scores, calibrated confidence scores, and feature-level explanation vectors for downstream components (C2–C4).

Table 1 – Functional Requirements

Nonfunctional requirements

(i). Performance: The system will be required to perform real-time inference (< 1 second per window) on consumer-grade hardware.

(ii). Accuracy: The classifier should achieve competitive accuracy and ROC-AUC comparable to the selected baseline model.

(iii). Calibration: Predicted probability must be well calibrated (low ECE).

(iv). Explainability: Explanations should clearly show the contribution of each feature to the stress decision.

(v). Privacy: The system shall run on de-identified data and shall make it possible to be run locally.

(vi). Usability: The output needs to be understandable by non-experts and adaptable to mobile or Web applications.

6. COMMERCIALIZATION PLAN

Target market and customer persona: The primary users are university students and working professionals who wish to monitor and manage stress levels. Secondary users include mental health practitioners who may use the system as a monitoring and support tool for stress-related conditions. In addition, wearable device manufacturers and digital health platforms may adopt and commercialize this technology by integrating the stress detection module into their products and services.

Value proposition: The proposed module enables continuous and unobtrusive stress monitoring using existing wearable devices. Unlike conventional emotion-based applications, the system provides calibrated stress probabilities and feature-level explanations, which improve the reliability and interpretability of predictions. These capabilities can increase user trust and support timely intervention strategies. Digital wellbeing platforms can leverage this module to support proactive stress management and enhance productivity.

Revenue model:

Possible options are

i. Licensing the algorithm to wearable device manufacturers and wellness application developers.
ii. Offering subscription-based access for institutions (e.g., universities and corporate wellness programs) to analyze aggregated stress trends while maintaining user privacy.
iii. Providing a premium analytics package with advanced reporting features and integration support for enterprise clients.

Cost estimation and pricing strategy: Since the solution relies on publicly available datasets and open-source tools (e.g., Python and SHAP), development costs are mainly associated with research effort and cloud computing resources (e.g., Google Colab Pro). A tiered pricing strategy could be adopted, offering basic functionality for free and advanced features through paid subscriptions.

Unique selling point: Most existing wellbeing applications provide general emotion-based feedback without clear explanations. In contrast, the proposed system produces calibrated stress probability scores along with interpretable feature-level explanations based on physiological signals. The use of subject-independent evaluation and probability calibration improves reliability and generalizability, making the system more suitable for real-world deployment.

Intellectual property issues: The algorithms developed in this project may have potential for patent protection. However, the use of public datasets and open-source libraries may limit the scope of intellectual property protection. Therefore, the research team should consider protecting unique model configurations, calibration techniques, and deployment pipelines, while potentially open-sourcing selected components to encourage wider adoption and collaboration.

7. BUDGET AND JUSTIFICATION

The project uses publicly available datasets and open-source software with direct costs being minimized.

Item	Cost Estimate	Justification
Google Colab Pro subscription (Optional)	USD 10 per month × 10 months ≈ USD 100 (LKR 30,000)	Gives the capability to access GPUs and train lightweight CNN-LSTM models and perform cross-validation; it is required to satisfy the performance requirements.
Computing hardware	Utilise existing laptops/workstations	No hardware-specific purchase is necessary since training is done on the cloud platforms, and currently the available devices would be sufficient for preprocessing and experimentation.
Software licenses	LKR 0	All tools used (Python, PyTorch/TensorFlow, SHAP) are open source.
Dataset acquisition	LKR 0	Datasets (WESAD and Kaggle wearable datasets) are publicly available.
Internet Cost	LKR 6,000	Covers connectivity for research and data downloads.
EDA sensor (future enhancement)	LKR 20,000	Optional cost to purchase a dedicated electrodermal activity sensor.
ECG sensor (future enhancement)	LKR 5,000	Optional cost to purchase an electrocardiogram sensor.
Total Estimated Budget	LKR 61,000	

Table 2 - Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

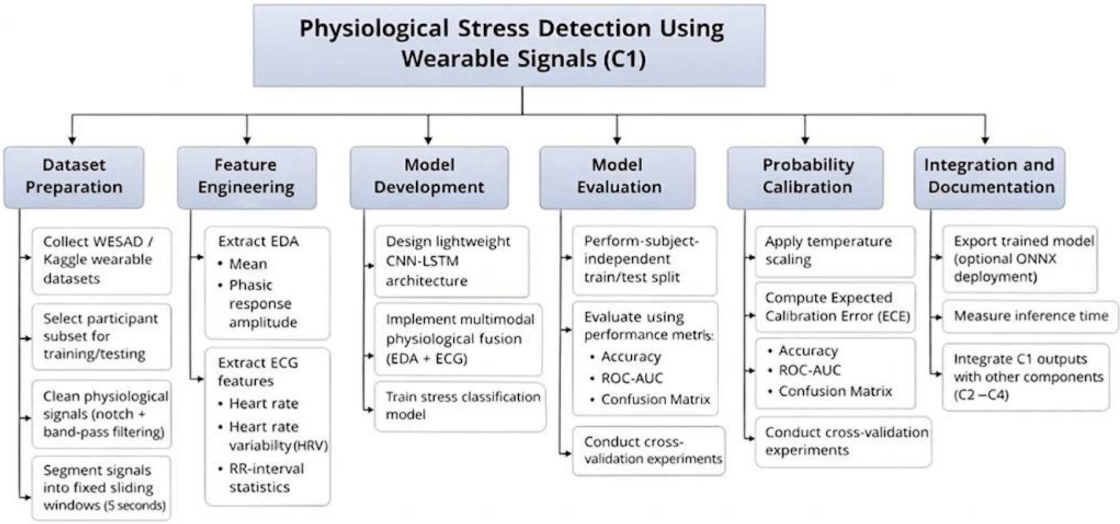

Figure 2 – Work Breakdown Structure

9. GANTT CHART

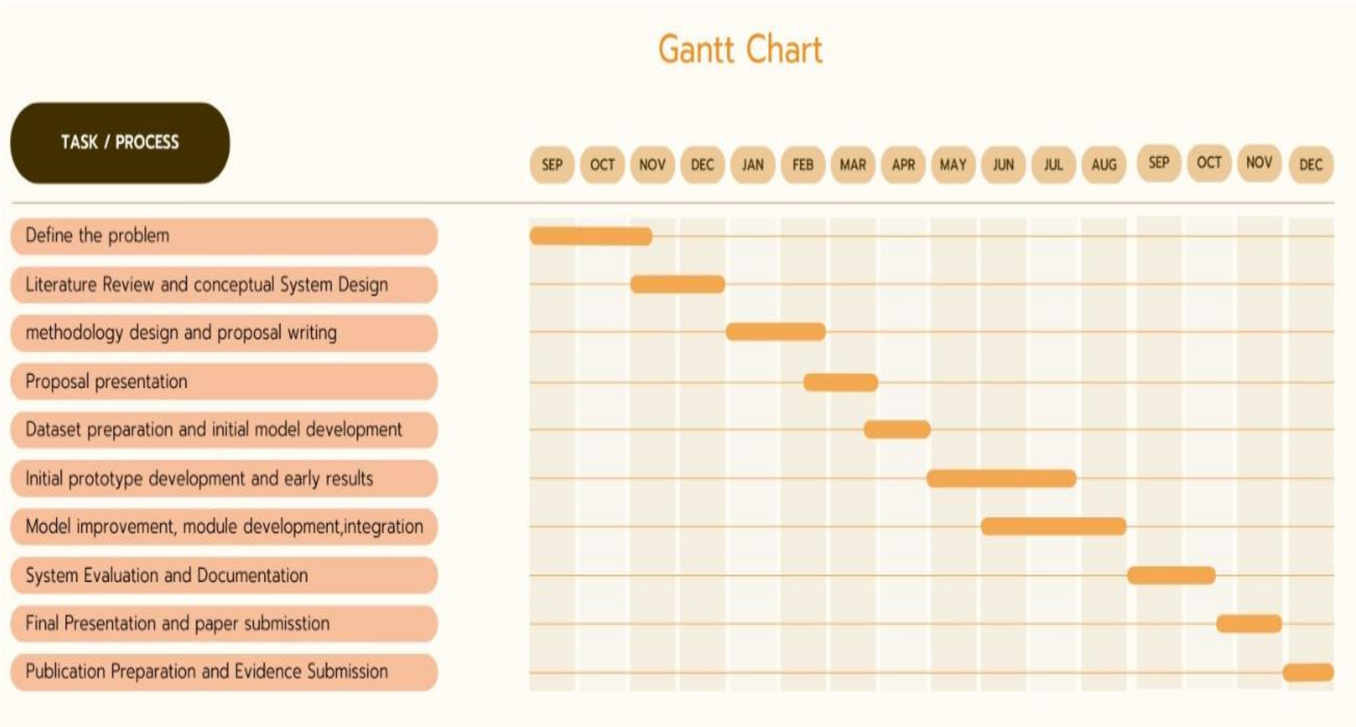

Figure 3 – Gantt Chart

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11. LIST OF APPENDICES

Risk Analysis Table

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R1	Limited dataset size or imbalance in the WESAD dataset may affect model training.	Medium	High	Use data balancing techniques and cross-validation. Experiment with different train/test splits.
R2	Physiological signals (EDA, ECG) may contain noise and artifacts affecting feature extraction.	High	Medium	Apply signal preprocessing such as filtering, noise removal, and normalization before feature extraction.
R3	Deep learning model (CNN-LSTM) may overfit due to small number of participants.	Medium	High	Use subject-independent validation, regularization techniques, and dropout layers.
R4	Model training may require high computational resources.	Medium	Medium	Use cloud resources such as Google Colab Pro for GPU acceleration.
R5	Model predictions may produce unreliable probabilities without calibration.	Medium	Medium	Apply probability calibration techniques such as temperature scaling and evaluate using Expected Calibration Error (ECE).
R6	Difficulty interpreting model predictions for end	Medium	Medium	Implement explainable AI

	users.			methods such as SHAP to show feature importance.
R7	Privacy concerns when dealing with physiological data.	Low	High	Use only publicly available de-identified datasets and avoid storing personal information.
R8	Integration issues between system components (C1–C4).	Low	High	Use standardized data outputs and APIs for communication between modules.

Table 3- Risk Analysis Table